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Question

- Interpretability often hopes each neuron corresponds to one human-understandable feature.
- Real models contain polysemantic neurons, suggesting features may not align with the neuron basis.
- When do limited dimensions force networks to pack features together?

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Geometry

- Features arrange as geometric packings—antipodal pairs, triangles, polygons and higher polytopes.
- Features have fractional effective dimensionality determined by angles and neighbor interference.
- Changing sparsity or importance causes phase transitions between monosemantic and superposed solutions.

2

Toy setup

- Inputs contain independent features with tunable importance and sparsity; only a subset is active.
- A linear bottleneck followed by ReLU reconstructs features with fewer hidden dimensions than inputs.
- $\hat{x} = \text{ReLU}(W^T W x + b)$

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Computation

FEATURES ARE DIRECTIONS

- The natural unit of representation is often a direction in activation space, not a neuron.
- Networks can compute with superposed features when nonlinear gating separates active combinations.
- Correlated or mutually exclusive features alter geometry and can share dimensions especially efficiently.

3

Core result

- When features are dense, the optimum dedicates orthogonal dimensions to important features.
- When features are sparse, non-orthogonal representations let many features share dimensions.
- ReLU removes much negative cross-talk; remaining interference is cheaper than dropping features.

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Interpretability

A MAP, NOT THE TERRITORY

- Polysemanticity can be a rational capacity trade-off rather than a training defect.
- Neuron-by-neuron analysis may miss distributed directions; sparse feature methods are better matched.
- These toy models omit real transformers' dependencies, hierarchical features and complex computation.

SUPERPOSITION

Toy Models of Superposition

How Networks Store More Features Than Dimensions | 2022

When useful features are sparse, a network can represent more features than it has dimensions by placing them in non-orthogonal directions; nonlinearities filter the resulting interference.

superposition

polysemanticity

interpretability